The mental status exam: describing what you see

The MSE is objective: just the facts, with quotes and pertinent negatives. Thought **process** is how the thoughts connect; thought **content** is what they are about.

Thought process	What it sounds like
Linear, goal-directed	A focused path from question to answer
Circumstantial	Unneeded detail and side remarks, **but reaches the answer**
Tangential	Each thought links to the last, but drifts away and **never returns** to the question
Flight of ideas	Fast, repeated jumps between loosely linked ideas, with pressured speech
Derailment (loose associations)	Jumps between unrelated ideas
Perseveration	Returns to the same answer or idea whatever is asked
Clanging	Words chosen for their **sound or rhyme**, not their meaning
Neologisms; word salad	Invented words; words strung together with no meaning
THOUGHT CONTENT (WHAT, NOT HOW)	
Ruminations	Unproductive mulling over the same unpleasant theme (depression, anxiety; not mania)
Delusions	Fixed false beliefs: persecutory, grandiose, referential, erotomanic, somatic, jealousy, guilt; thought insertion, withdrawal, broadcasting. In mood episodes they match the mood
Delusional syndromes	**Capgras**: a loved one replaced by an imposter. **Cotard**: one is dead or rotting. **Fregoli**: a stranger is someone familiar in disguise

Perception and dissociation

Illusion vs hallucination	Misreading a real stimulus vs perceiving with no stimulus
Hypnagogic; hypnopompic	Brief perceptions falling asleep; waking up. Not psychosis
Depersonalization	Feeling unreal or **outside one's own body**, watching oneself
Derealization	The world feels unreal
Fugue; jamais vu; déjà vu	Amnesia for identity, often with travel; the familiar feels new; the new feels familiar

Mood vs affect: mood is the patient's own word; affect is what you observe (range, reactivity, congruence).

Speech and movement

	Depression	Mania	Anxiety
Speech	Slow, quiet, long pauses; poverty of speech	Loud, rapid, **pressured**; hard to interrupt	Rapid but interruptible
Psychomotor	**Slowing**	Agitation	**Agitation**: fidgeting, pacing, edge of the chair; hypervigilant

Movement finding	Points to
Waxy flexibility, posturing, echolalia, echopraxia, negativism	Catatonia
Cogwheeling, tremor, shuffling	Parkinsonism (incl. from antipsychotics)
Lip smacking, grimacing, blinking	Tardive dyskinesia (long-term antipsychotics)
Torticollis, oculogyric crisis	Dystonia

For thought process, ask: **"Did the answer get back to my question?"** Yes, slowly → circumstantial. Never → tangential. Linked by sound → clanging. Fast and loosely linked with pressured speech → flight of ideas. No logical link → derailment.

Sources: E25 Mental Status Exam slides (faculty version) and notes; Exam Review deck.

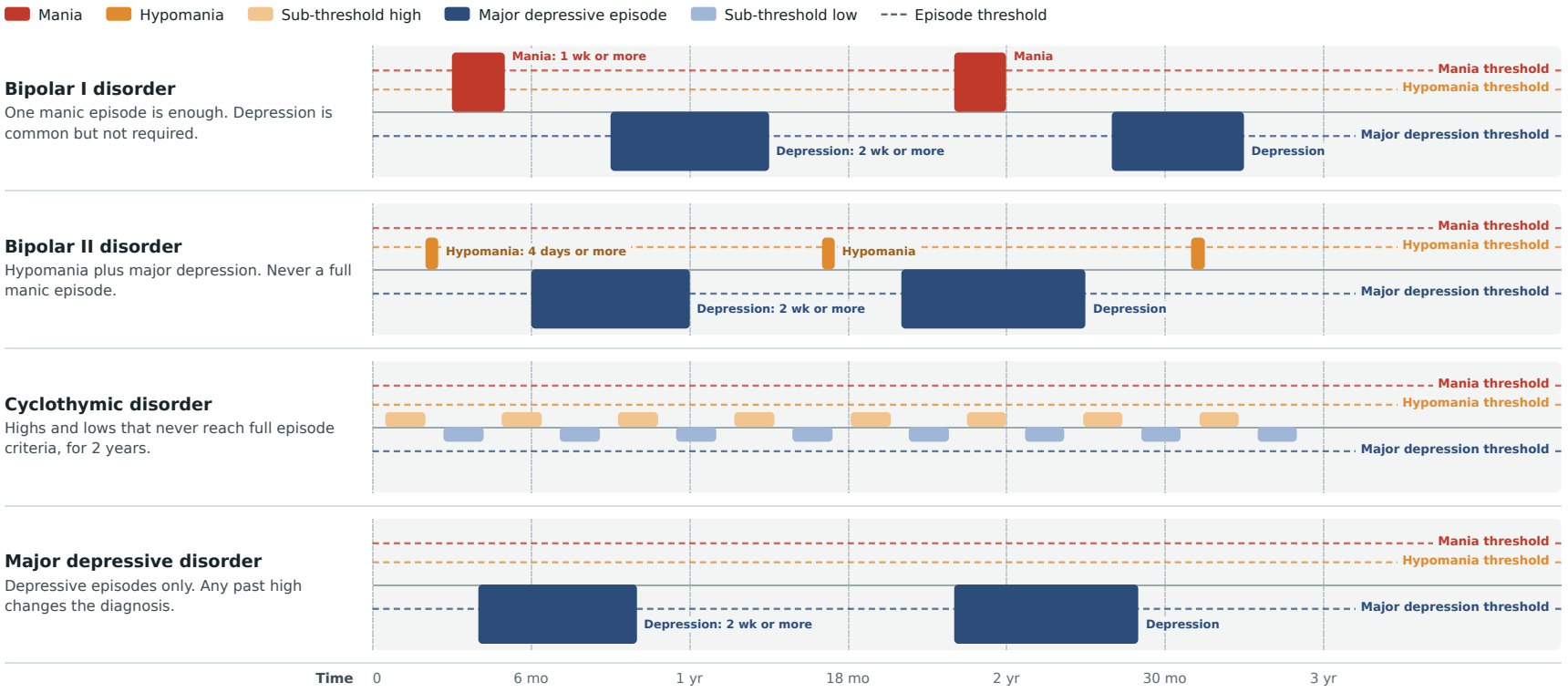
Bipolar I, bipolar II, cyclothymia and depression

These disorders are sorted by the most severe high the patient has *ever* had. Severity of the high, not how many depressions came before, decides the diagnosis.

Diagnosis	Episodes required	Episodes that rule it out	Typical course	First-line treatment
Bipolar I disorder	At least one **manic** episode	None: one manic episode, ever, is enough	Mania, often with depressive episodes (not required)	Lithium, valproate or a second-generation antipsychotic; avoid antidepressant monotherapy
Bipolar II disorder	At least one **hypomanic** and one **major depressive** episode	Any manic episode (makes it bipolar I)	Mostly depression; brief hypomanic periods	Lithium, lamotrigine or quetiapine; psychotherapy
Cyclothymic disorder	2 years of hypomanic *and* depressive symptoms, present half the time or more	Any full major depressive, manic or hypomanic episode	Frequent, sub-threshold highs and lows	Psychotherapy; mood stabilizer if impairing
Major depressive disorder	At least one major depressive episode	Any manic or hypomanic episode, ever	Depressive episodes with return to baseline	SSRI or SNRI, psychotherapy; ECT if severe or psychotic

Picture it as a mood chart

Above the solid line is elevated mood, below is depressed mood; height shows severity. Bars are drawn to scale over 3 years at typical untreated lengths (mania weeks to months, depression around 6 months, hypomania days to weeks). Labels give the minimum duration that counts.



Mania vs. hypomania

	Mania	Hypomania
Duration	1 week or more (any length if hospitalized)	4 consecutive days or more
Symptoms	3 or more (4 if mood is only irritable): decreased need for sleep, pressured speech, racing thoughts or flight of ideas, grandiosity, distractibility, more goal-directed activity, risky behavior	Same symptom list and count
Severity	**Marked impairment, psychosis or hospitalization**	A clear change others notice, without marked impairment, psychosis or hospitalization
Diagnosis it points to	Bipolar I disorder	Bipolar II disorder (with a major depressive episode)

Ask about the **lifetime** history of highs before calling any depression unipolar. Then ask: **did the high cause marked impairment, psychosis or hospitalization?** Yes → mania → **bipolar I**. No, but it lasted 4 days or more → hypomania → **bipolar II**. Never met either for 2 years of ups and downs → **cyclothymia**.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

MBB PSYCHIATRY · MOOD DISORDERS

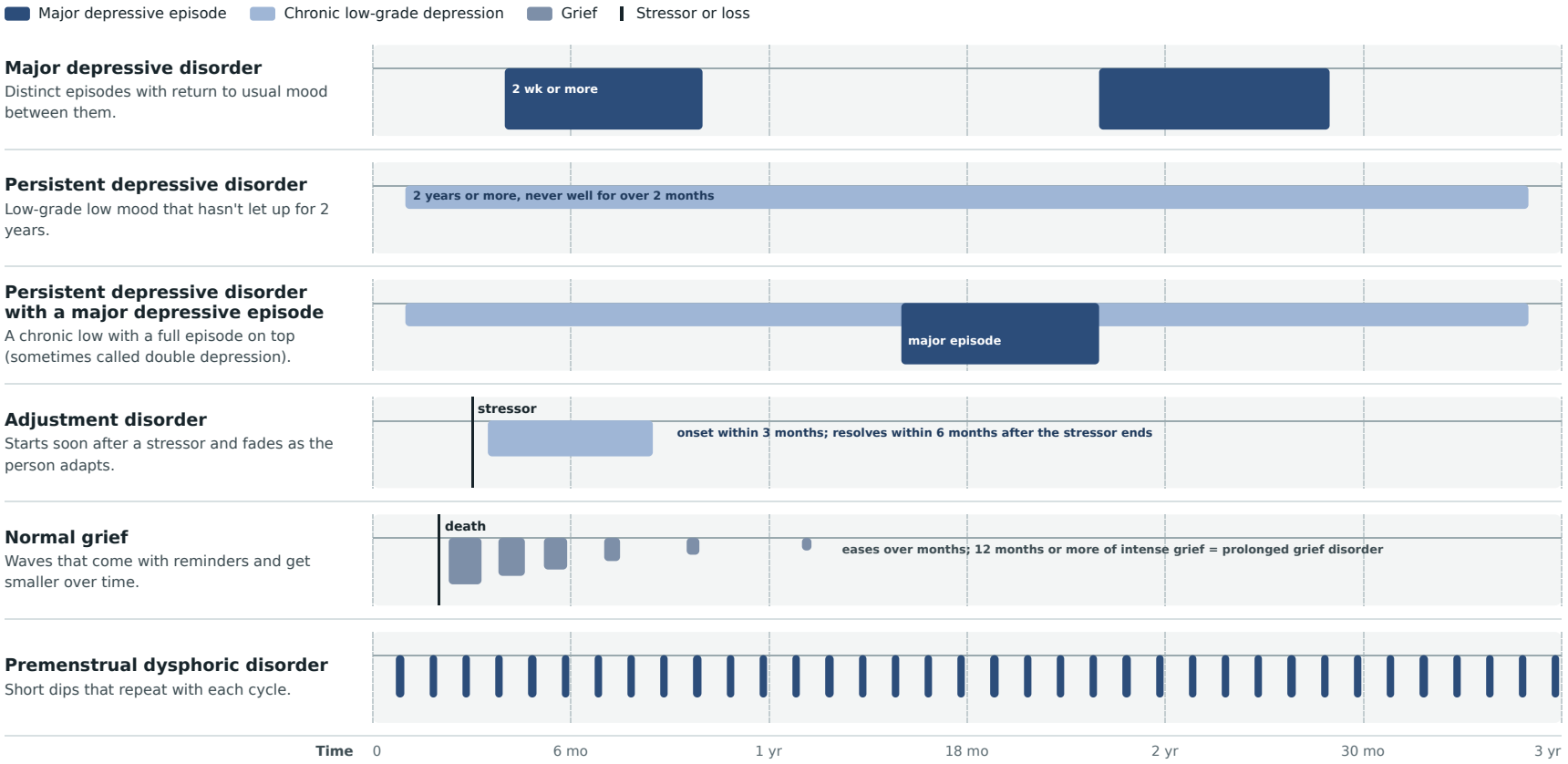
Telling the depressive disorders apart

Most low-mood presentations are sorted by duration and by what the mood is tied to. Rule out bipolar disorder by asking about past highs.

Diagnosis	Timing	What it looks like	Key discriminator	First-line treatment
Major depressive disorder	Episodes of 2 weeks or more	5 of 9 symptoms including low mood or loss of interest	Pervasive, with neurovegetative symptoms and impairment	SSRI or SNRI, psychotherapy (CBT, behavioral activation, IPT); ECT if severe or psychotic
Persistent depressive disorder	**2 years or more**, never well for over 2 months	Low mood most days plus 2 of: appetite, sleep, energy, self-esteem, concentration, hopelessness	Chronic and lower-grade; a major depressive episode can occur on top	SSRI plus psychotherapy
Adjustment disorder	Starts **within 3 months of a stressor**; ends within 6 months after it	Distress out of proportion to the stressor, or impairment	Doesn't meet criteria for another disorder	Psychotherapy
Normal grief	After a death; eases over time	Waves of sadness tied to reminders	Self-esteem intact; humor and pleasure still possible	Support
Prolonged grief disorder	**12 months or more** after the death	Intense yearning or preoccupation with the deceased most days	Persistent, impairing grief that exceeds cultural norms	Grief-focused psychotherapy
Premenstrual dysphoric disorder	Week before menses, in most cycles over a year	5 or more mood and physical symptoms	Improves after menses starts; minimal the week after	SSRI (continuous or luteal phase); some oral contraceptives
Substance- or medical-induced depressive disorder	Follows a drug, withdrawal or illness	Depressive syndrome	Timing with the cause (steroids, interferon, hypothyroidism, stroke, others)	Treat the cause

Picture it on a timeline

The line is the person's usual mood. Bars below it show how deep and how long the low mood runs, drawn to scale over 3 years. PMDD dips are about a week each month.



Ask **how long** it has lasted and **what it is tied to**. A stressor or a death points to adjustment disorder or grief; the menstrual cycle points to PMDD; 2 years without relief points to persistent depressive disorder. A full 2-week episode can be major depression even when one of these is also present.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

MBB PSYCHIATRY · MOOD DISORDERS · PHARMACOLOGY

The antidepressants

SSRIs are first line for major depression and most anxiety disorders. Every class takes weeks to work, so choose by side effects and by the patient's other problems.

Class	Drugs	Mechanism	Best used for	Watch for
SSRI	Fluoxetine, sertraline, paroxetine, citalopram, escitalopram, fluvoxamine	Selectively inhibit serotonin reuptake at the presynaptic pump	First line for major depression and anxiety disorders, including PTSD and OCD (higher doses for OCD)	GI upset, sexual dysfunction; **suicidal thinking up to age 24** (all SSRIs); citalopram prolongs QTc
SNRI	Venlafaxine, duloxetine	Serotonin and norepinephrine reuptake inhibition	Major depression and anxiety disorders; duloxetine also for peripheral neuropathy and fibromyalgia	GI upset, sexual dysfunction, sweating and heat intolerance, **raised blood pressure**
NDRI	Bupropion	Norepinephrine and dopamine reuptake inhibition	Major depression when sexual side effects must be avoided	Activating, insomnia; **lowers the seizure threshold**
α2 antagonist	Mirtazapine	Blocks α2 receptors, raising norepinephrine and serotonin activity; potent antihistamine	Major depression, especially with insomnia or poor appetite; PTSD	Very sedating, weight gain
SARI	Trazodone	Serotonin reuptake inhibition plus 5-HT2 antagonism	Sleep aid; major depression	Sedation, **priapism**
Tricyclic (TCA)	Amitriptyline, nortriptyline, imipramine, desipramine; clomipramine for OCD	Serotonin and norepinephrine reuptake inhibition; also antihistamine, antimuscarinic and anti-α1	Major depression, anxiety disorders, neuropathy and chronic pain	Sedation, weight gain, constipation, orthostasis; **lethal in overdose**: QRS widening, heart block, hypotension
MAOI	Phenelzine, tranylcypromine, selegiline	Inhibit monoamine oxidase, so serotonin, norepinephrine and dopamine are broken down less	Treatment-resistant depression; anxiety disorders	**Hypertensive crisis** with tyramine foods or stimulants (diet must be restricted); serotonin syndrome with serotonergic drugs

When the first antidepressants don't work

Treatment-resistant depression usually means two adequate antidepressant trials have failed.

MAXIMIZE FIRST

Medications and therapy

- Augment: lithium, MAOIs, clozapine
- CBT or other psychotherapy, light therapy, exercise

GOLD STANDARD

ECT

- **First line for psychotic depression**
- Severe depression, or depression that doesn't respond to medications
- Catatonia and neuroleptic malignant syndrome
- Main risk: cognitive effects

IF ECT IS DECLINED OR UNSAFE

Ketamine and esketamine

- Ketamine infusions: used but not FDA-approved
- Esketamine: FDA-approved, given with an oral antidepressant
- Monitor vitals and ECG; dissociation, hallucinations, psychosis

MODERATE DEPRESSION

rTMS

- Referral option alongside esketamine
- Treatments can be combined

EMERGENCY

Serotonin syndrome

- Several serotonergic drugs, within ~48 h of a change
- Tremor, sweating, hypertension, **hyperreflexia, clonus**
- Stop drugs; cyproheptadine, benzodiazepines

Ask: **"What else does this patient need the drug to do, or not do?"** Neuropathic pain → duloxetine. Sexual side effects → bupropion. Insomnia or weight loss → mirtazapine. Seizures → avoid bupropion. Overdose risk → avoid tricyclics. Bipolar history → no antidepressant alone. Psychotic or refractory depression → ECT.

Sources: E25 Mood Disorders and Pharmacology slides; Psychiatry Exam Review deck (treatment-resistant depression, serotonin syndrome).

MBB PSYCHIATRY · MOOD DISORDERS · PHARMACOLOGY

Mood stabilizers for bipolar disorder

Pick by phase: what treats acute mania, and what prevents the next episode. Lithium is the gold standard; the adverse effects of the anticonvulsants are high-yield.

Drug	Acute mania	Maintenance	Mechanism	Common adverse effects	Rare but serious
Lithium gold standard	Yes	Yes	Unclear: second-messenger pathways (adenylyl cyclase, phosphoinositide, protein kinase C), neuroprotective proteins such as BDNF	Short term: nausea, tremor. Long term: **thyroid and renal dysfunction**	**Narrow therapeutic index** with a severe toxidrome (below)
Valproic acid	Yes		Anticonvulsant	Weight gain, PCOS, transaminitis	Pancreatitis, **fetal neural tube defects**
Carbamazepine	Yes		Anticonvulsant; induces CYP450 3A4	Numerous CYP450 drug interactions	**Agranulocytosis**, Stevens-Johnson syndrome
Lamotrigine		Yes	Anticonvulsant	Doses must be raised and lowered very gradually	**Stevens-Johnson syndrome / TEN**
Second-generation antipsychotics	Yes	Yes	Risperidone, olanzapine, quetiapine and others	Weight gain and metabolic effects; see the antipsychotics page	Neuroleptic malignant syndrome

Lithium toxicity climbs with the level

Acute toxicity, from the mood disorders lecture (for reference).

MILD	MODERATE	SEVERE
Lethargy, weakness, nausea	Vision changes, nystagmus, confusion, ataxia, hyperreflexia	Clonus, arrhythmias, seizures, coma or death

Match the drug to the diagnosis

Diagnosis	First-line medication
Bipolar I disorder	Lithium, valproate or a second-generation antipsychotic; **avoid antidepressant monotherapy**
Bipolar II disorder	Lithium, lamotrigine or quetiapine; psychotherapy
Cyclothymic disorder	Psychotherapy; a mood stabilizer if impairing

Board favorites

One adverse effect per drug.

Lithium
Hypothyroidism, renal dysfunction, narrow window

Valproate
Neural tube defects, pancreatitis, liver

Carbamazepine
Agranulocytosis, CYP3A4 induction

Lamotrigine
Stevens-Johnson syndrome; titrate slowly

Second-generation antipsychotics
Weight gain, diabetes, dyslipidemia

Any antidepressant alone
Can switch bipolar depression into mania

Ask: **“Is this acute mania or prevention?”** Acute mania → lithium, valproate, carbamazepine or an antipsychotic. Maintenance → lithium, lamotrigine or a second-generation antipsychotic. Pregnancy → avoid valproate.

Mania after an antidepressant: a full manic episode (psychosis, impairment or hospitalization) that follows an antidepressant suggests the real diagnosis is bipolar I. **Stop the antidepressant and start a mood stabilizer such as lithium.**

Grief or major depression? Sadness, insomnia, fatigue and poor appetite are common in grief. **Worthlessness**, a suicide plan, marked impairment, psychomotor slowing or psychosis point to major depression.

Sources: E25 Mood Disorders and Pharmacology slides (lithium, anticonvulsant table, second-generation antipsychotics); first-line choices as on the “Bipolar I, bipolar II, cyclothymia” sheet; Exam Review deck.

MBB PSYCHIATRY · ANXIETY DISORDERS

Telling the anxiety disorders apart

The anxiety disorders share the same physical symptoms. What separates them is what the fear is about and what the person does to cope.

Disorder	What the fear is about	What the person does	Duration	First-line treatment	Tell it from
Generalized anxiety disorder	Many everyday matters: work, health, money, family	Restless, tense, tired, poor sleep and concentration	6 months or more	SSRI or SNRI, CBT; buspirone as an add-on	Panic disorder: discrete attacks rather than constant worry
Panic disorder	The next attack and what it means ("I'm dying", "losing control")	Changes behavior to avoid attacks	1 month of worry after recurrent **unexpected** attacks	SSRI, CBT	Cardiac, thyroid or stimulant causes
Agoraphobia	Places where escape or help is hard: transit, open or enclosed spaces, crowds, being out alone	Avoids them or needs a companion	6 months or more	Exposure-based CBT, SSRI	Social anxiety: fear of judgment, not of being trapped
Social anxiety disorder	Being judged or embarrassed by others	Avoids or endures social or performance situations	6 months or more	CBT, SSRI; propranolol as needed for performance-only type	Avoidant personality: pervasive inadequacy across life
Specific phobia	One object or situation (heights, animals, needles, flying)	Immediate fear on exposure, avoidance	6 months or more	Exposure therapy	Agoraphobia: two or more situation types
Separation anxiety disorder	Harm to or separation from attachment figures	Refuses to be apart, nightmares, physical complaints	4 weeks in children, 6 months in adults	CBT with exposure	Agoraphobia: fear of the place, not of separation

Panic attack or panic disorder?

A SYMPTOM

Panic attack

- Abrupt surge of fear peaking within minutes
- 4 or more physical or cognitive symptoms (palpitations, sweating, trembling, shortness of breath, chest pain, dizziness, fear of dying)
- Can occur in any anxiety disorder, depression or PTSD

A DISORDER

Panic disorder

- Recurrent attacks that come **out of the blue**
- 1 month of worry about more attacks or behavior change to prevent them
- Not explained by a substance or medical condition

EXPECTED ATTACKS

Attacks with a trigger

- Only in social situations → social anxiety disorder
- Only with one object → specific phobia
- Only with trauma reminders → PTSD
- Only in hard-to-escape places → agoraphobia

Ask the patient: **"What exactly are you afraid will happen?"** Many things → GAD. The next attack → panic disorder. Being trapped → agoraphobia. Being judged → social anxiety. One thing → specific phobia. Losing someone → separation anxiety.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

OCD and its relatives

These disorders pair a thought or urge with a repetitive behavior. Name the thought, name the behavior, and ask how the person feels about it.

Disorder	The thought or urge	The repetitive behavior	Insight and attitude	First-line treatment
Obsessive-compulsive disorder	Intrusive, unwanted obsessions (contamination, harm, symmetry, taboo thoughts)	Compulsions to neutralize them: washing, checking, counting, mental rituals	Usually sees them as excessive; distressed by them	Exposure and response prevention; SSRI at higher doses; clomipramine
Body dysmorphic disorder	A perceived flaw in appearance others can't see or find slight	Mirror checking, grooming, skin picking, reassurance seeking, cosmetic procedures	Insight often poor	CBT with exposure and response prevention; SSRI
Hoarding disorder	Strong need to save; distress at discarding	Accumulating possessions until living space is unusable	Insight often poor	CBT for hoarding
Trichotillomania	Rising tension before pulling	Hair pulling with hair loss	Wants to stop; relief or pleasure after pulling	Habit reversal therapy
Excoriation disorder	Rising tension before picking	Skin picking with lesions	Wants to stop; relief after picking	Habit reversal therapy
FOR COMPARISON				
Obsessive-compulsive personality disorder	No intrusive thoughts	Perfectionism, rules, lists, control, can't delegate	**Sees the behavior as right**; others are frustrated	Psychotherapy
Illness anxiety disorder	Fear of having a serious illness	Body checking, reassurance seeking or avoiding doctors	Few or no symptoms	CBT

Obsession, worry or delusion?

GAD

Worry

- About real-life matters
- Feels reasonable to the patient, just too much
- No ritual to neutralize it

OCD

Obsession

- Intrusive and unwanted
- Usually recognized as excessive (insight good, poor or absent)
- Followed by a compulsion

PSYCHOSIS

Delusion

- Fixed false belief
- Held with full conviction despite evidence
- No sense that it is excessive

Ask: **"Does this thought make sense to you, and do you want to stop?"** Unwanted thought plus a ritual → OCD. Appearance focus → body dysmorphic disorder. Tension relieved by pulling or picking → habit reversal territory. Behavior the person thinks is correct → OCPD.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

MBB PSYCHIATRY · TRAUMA- AND STRESSOR-RELATED DISORDERS

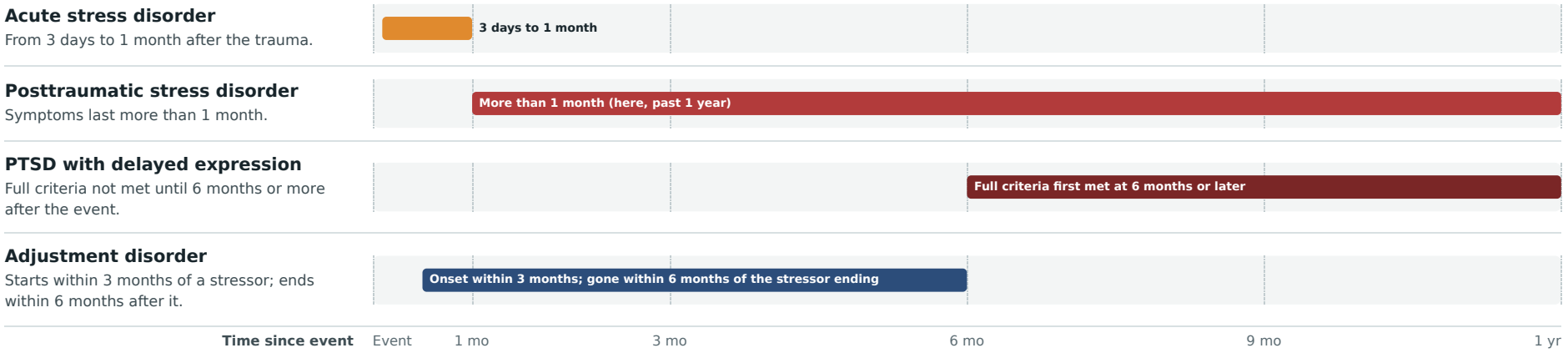
Acute stress, PTSD and adjustment disorder

Two questions separate these: was the event a trauma or an ordinary stressor, and how long has it been since?

Diagnosis	What happened	Timing	Core symptoms	First-line treatment
Acute stress disorder	**Trauma**: actual or threatened death, serious injury or sexual violence, experienced or witnessed	3 days to 1 month after the event	9 or more symptoms of intrusion, negative mood, dissociation, avoidance and arousal	Trauma-focused CBT
Posttraumatic stress disorder	**Trauma**, as above	**More than 1 month**; delayed expression if full criteria start 6 months or more later	Intrusion, avoidance, negative thoughts and mood, hyperarousal	Prolonged exposure or cognitive processing therapy; SSRI or SNRI; prazosin for nightmares
Adjustment disorder	Any **ordinary stressor**: divorce, job loss, illness, a move	Starts within 3 months; resolves within 6 months after the stressor ends	Distress or impairment out of proportion; no re-experiencing	Psychotherapy
Prolonged grief disorder	Death of someone close	12 months or more after the death	Intense yearning or preoccupation, identity disruption, emotional pain	Grief-focused psychotherapy

Picture it on a timeline

Drawn to scale over the first year after the event.



PTSD needs all four clusters: intrusion (1 or more), avoidance (1 or more), negative thoughts and mood (2 or more), and arousal (2 or more).

Trauma or stressor? PTSD and acute stress disorder require a life-threatening or violent event. A divorce or job loss points to adjustment disorder.

Acute stress disorder can become PTSD. If symptoms continue past 1 month, change the diagnosis.

Avoid benzodiazepines in PTSD. They don't treat the core symptoms and may interfere with recovery.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

Medications for anxiety, OCD and trauma

Therapy is the gold standard for every anxiety disorder; medication adds to it. Separate the daily drug that prevents symptoms from the as-needed one for the moment.

Disorder	Daily, long-term medication	As needed or add-on	Avoid or not helpful
Generalized anxiety disorder	SSRIs or SNRIs first line; TCAs and MAOIs also work	Buspirone, usually added to an SSRI or SNRI	Caffeine
Panic disorder	SSRIs, SNRIs, TCAs, MAOIs (4+ weeks to work)	Benzodiazepine or hydroxyzine, briefly, while the SSRI starts working	Benzodiazepines as a long-term strategy
Agoraphobia	SSRIs, SNRIs, TCAs, at higher doses than for mood disorders	Benzodiazepines short term; hydroxyzine as needed	
Social anxiety disorder	SSRI (with CBT)	Propranolol as needed for the **performance-only** type (tests, speeches)	Propranolol for generalized social anxiety
Specific phobia	Medications generally don’t help; treat with exposure therapy	Benzodiazepines sometimes, for an acute episode	
TRAUMA- AND STRESSOR-RELATED			
PTSD	Sertraline, paroxetine (FDA-approved); fluoxetine, venlafaxine, duloxetine, mirtazapine	**Prazosin for nightmares**	**Benzodiazepines**: no benefit, may block recovery
OBSESSIVE-COMPULSIVE AND RELATED			
OCD	SSRIs, including fluvoxamine, at **higher doses for longer**; clomipramine	Treatment-resistant: deep brain stimulation, TMS, psychosurgery	
Body dysmorphic; hoarding	BDD: SSRIs with CBT. Hoarding: SSRIs help mainly if OCD symptoms are present		
Trichotillomania, excoriation	SSRIs help only coexisting anxiety or depression	Off-label, under study: N-acetylcysteine, naltrexone	

The as-needed and add-on drugs

Buspirone

5-HT1A partial agonist

- GAD, mostly as augmentation
- 1–2+ weeks to work; 2–3 doses a day
- Well tolerated
- **No misuse potential or CNS depression**

Benzodiazepines

Enhance GABA at the GABA-A receptor

- Withdrawal states; short-term anxiety; sleep
- Sedation, falls; respiratory depression with alcohol or opioids
- Misuse, dependence, **withdrawal seizures**
- LOT (lorazepam, oxazepam, temazepam): safest in liver disease

Hydroxyzine

H1 antihistamine

- Breakthrough anxiety; anxiety-related insomnia
- Sedation; anticholinergic effects, confusion in older patients

Propranolol

Non-selective β-blocker

- As needed for performance or test anxiety
- Bronchospasm, chest tightness, hypotension, bradycardia

Prazosin

α1 antagonist

- PTSD-related nightmares
- Orthostasis, light-headedness, hypotension, palpitations, headache

Ask: **“Is this for every day, or for the moment?”** Every day → SSRI or SNRI (weeks to work). Panic while it starts → brief benzodiazepine or hydroxyzine. A speech or test → propranolol. Nightmares → prazosin.

Sources: E25 Anxiety Disorders and Pharmacology slides (faculty version) and speaker notes.

MBB PSYCHIATRY · DIFFERENTIAL DIAGNOSIS

Telling the "schizo" disorders apart

Ask three questions: is there frank psychosis, how long has it lasted, and where do mood episodes fall? The two personality disorders have no sustained psychosis. Brief psychotic and delusional disorder are shown as the closest neighbors.

Disorder	Frank psychosis?	Duration	Mood episodes	Function and social pattern	Tell-tale sign
PERSONALITY DISORDERS (CLUSTER A)					
Schizoid personality disorder	**No**	Lifelong pattern, from early adulthood	Not part of the picture	Detached, prefers to be alone, restricted emotion, indifferent to praise or criticism	Content to be alone; no odd beliefs
Schizotypal personality disorder	**No (brief, transient episodes at most)**	Lifelong pattern, from early adulthood	Not part of the picture	Few close relationships; social anxiety that doesn't ease with familiarity	Odd beliefs, magical thinking, ideas of reference, eccentric speech or dress
PSYCHOTIC DISORDERS					
Brief psychotic disorder	**Yes, 1 or more symptoms**	1 day to under 1 month	Not required	Full return to prior functioning	Sudden onset, often after a major stressor or postpartum
Schizophreniform disorder	**Yes, 2 or more symptoms**	1 to under 6 months	Absent or brief	Decline not required	Schizophrenia picture that hasn't reached 6 months
Schizophrenia	**Yes, 2 or more symptoms**	6 months or more, including 1 month of active symptoms	Absent or brief relative to the whole illness	Clear decline in work, relationships or self-care	Duration plus functional decline; negative symptoms common
Schizoaffective disorder	**Yes**	Lifetime illness	Major mood episodes for most of the illness	Varies	Psychosis for 2 weeks or more with no mood episode, and mood episodes for most of the illness
Delusional disorder	**Delusions only**	1 month or more	Brief if present	Functioning and behavior otherwise not obviously odd	Non-bizarre delusion; no prominent hallucinations, disorganization or negative symptoms



Schizoid keeps others away and is content alone. **Schizotypal** is odd and magical. Neither has sustained psychosis.

Counted symptoms: delusions, hallucinations, disorganized speech, disorganized or catatonic behavior, negative symptoms. For schizophreniform and schizophrenia, at least one must be delusions, hallucinations or disorganized speech.

Schizoaffective adds the affective (mood) part: psychosis without mood symptoms for 2 weeks or more *and* mood episodes for most of the illness. Psychosis only during mood episodes is a mood disorder with psychotic features.

Schizoid vs. avoidant: both are isolated, but the avoidant person wants relationships and stays away for fear of rejection; the schizoid person doesn't want them.

Rule out substances, medications, delirium and medical causes before any primary psychotic diagnosis. Durations follow DSM-5-TR for adults.

MBB PSYCHIATRY · PSYCHOTIC DISORDERS

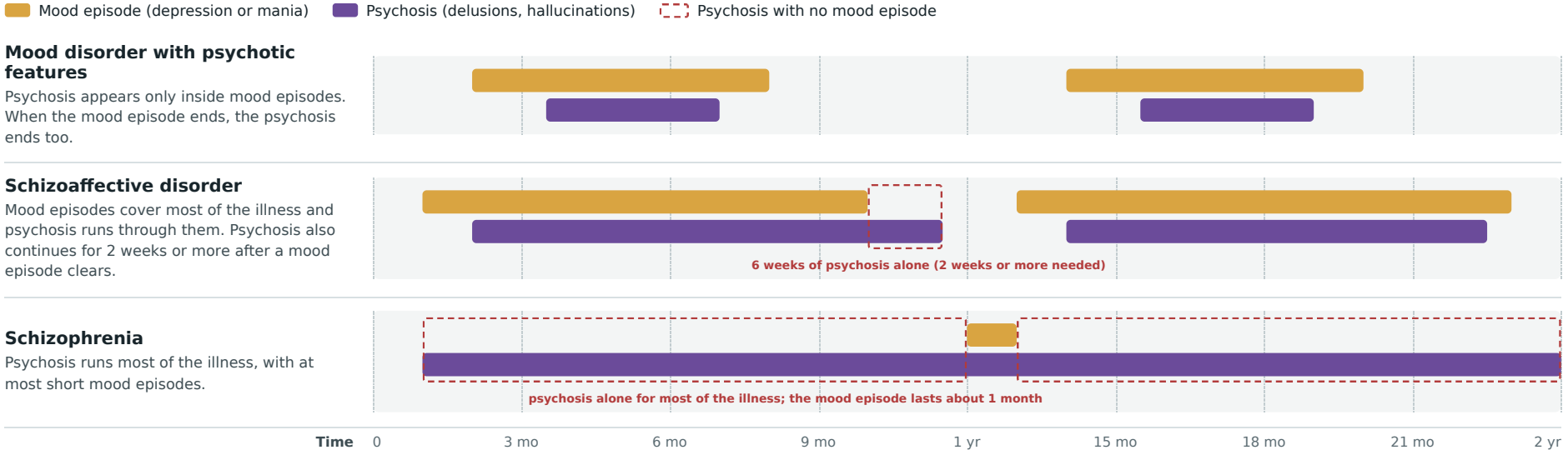
Psychosis and mood: three related diagnoses

Psychosis and mood episodes can overlap in all three. What separates them is whether psychosis ever happens without a mood episode, and how much of the illness the mood episodes fill.

Diagnosis	Psychosis during mood episodes?	Psychosis without a mood episode?	Mood episodes for most of the illness?
Mood disorder with psychotic features	Yes	**Never**	Yes
Schizoaffective disorder	Yes	**Yes, 2 weeks or more**	**Yes**
Schizophrenia	Can happen	Yes, most of the time	**No, they're brief**

Picture it on a timeline

Each bar is 2 years of illness, drawn to scale. Mood episodes are shown at typical untreated lengths of about 6 months.



Draw the timeline and ask: **"When the mood was normal, was the patient still psychotic?"** Never → **mood disorder with psychotic features**. Yes, 2 weeks or more, and mood episodes fill most of the illness → **schizoaffective disorder**. Yes, and mood episodes are brief → **schizophrenia**.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

MBB PSYCHIATRY · PSYCHOTIC DISORDERS

Is the psychosis primary or secondary?

Psychosis is a syndrome, not a diagnosis. The first job is to rule out substances, medications, delirium and medical illness.

Cause	Onset	Awareness and attention	Hallucinations	Other clues	What resolves it
Schizophrenia spectrum	Gradual, usually late teens to early 30s	Clear; oriented	Mostly auditory	Prodrome of withdrawal and decline; negative symptoms	Antipsychotics; often chronic
Mood disorder with psychotic features	With the mood episode	Clear	Auditory; themes match the mood (guilt, ruin, grandiosity)	Full depressive or manic episode	Treating the mood episode; ECT if severe
Substance- or medication-induced	**Abrupt**, tied to use or withdrawal	Can be impaired	Visual or tactile common	Dilated or pinpoint pupils, abnormal vital signs, positive toxicology	Clears as the substance clears
Psychosis due to a medical condition	Follows the illness	Often impaired	**Visual, olfactory or tactile**	Older first onset, neurologic signs, seizures, endocrine or autoimmune disease	Treating the condition
Delirium	**Hours to days**	**Inattentive, fluctuating**	Visual common	Hospitalized or older, infection, new drugs, withdrawal	Finding and treating the cause
Dementia with Lewy bodies	Gradual	Fluctuating	Well-formed visual, early	Parkinsonism, sleep behavior disorder, sensitivity to antipsychotics	Not reversible; cholinesterase inhibitors help

Red flags for a secondary cause

- **Onset after 40,** or a first episode that comes on over hours to days.
- **Visual, tactile or olfactory hallucinations** rather than voices.
- **Fluctuating attention or awareness,** disorientation, or a change in level of consciousness.
- **Abnormal vital signs, pupils or neurologic exam,** a new medication, or recent substance use or withdrawal.

Before naming a primary psychotic disorder, ask: **could a substance, a medication, delirium or a medical illness explain this?** Check vital signs, pupils, attention, toxicology and recent medication changes. Only when the answer is no does the timeline of psychosis and mood decide the diagnosis.

Durations and thresholds follow DSM-5-TR for adults. Rule out substances, medications and medical causes first.

MBB PSYCHIATRY · PSYCHOTIC DISORDERS · PHARMACOLOGY

Antipsychotics

All of them block dopamine D2 receptors, or partly activate them. Which pathway is blocked explains both the benefit and most side effects.

Group	Mechanism	Examples	Side-effect tilt
TYPICAL (FIRST GENERATION)			
High potency	Strong D2 antagonism	Haloperidol, fluphenazine, trifluoperazine, perphenazine, pimozide	**More EPS**, less anticholinergic
Low potency	Weaker D2 antagonism	Chlorpromazine, thioridazine	Less EPS; more anticholinergic and anti-α1
ATYPICAL (SECOND GENERATION)			
D2 + 5-HT2A antagonists	5-HT2A blockade added (thought to lower EPS)	Risperidone, olanzapine, quetiapine, ziprasidone, clozapine	**Metabolic**: weight, diabetes, lipids
D2 partial agonists	Partial agonism at D2	Aripiprazole, cariprazine	Akathisia
Muscarinic: xanomeline + trospium (Cobenfy)	M1/M4 agonist lowers dopamine indirectly; trospium blocks peripheral effects		GI effects; less metabolic and EPS risk

Dopamine pathway	Runs from → to	Blocking D2 here causes
Mesolimbic	Ventral tegmentum → nucleus accumbens	Fewer delusions and hallucinations (**the target**)
Mesocortical	Ventral tegmentum → prefrontal cortex	Worse negative symptoms (cognition, motivation)
Nigrostriatal	Substantia nigra → basal ganglia	**EPS**: dystonia, akathisia, parkinsonism; tardive dyskinesia. A relative acetylcholine excess, so weakly anticholinergic drugs cause the most
Tuberoinfundibular	Hypothalamus → anterior pituitary	Raised prolactin: galactorrhea, sexual dysfunction

Which drug is worst for…

Weight, lipids, diabetes
Clozapine, olanzapine (lower risk: lurasidone, ziprasidone, aripiprazole)

EPS and prolactin
Risperidone (and high-potency typicals)

Akathisia
Aripiprazole

Agranulocytosis
Clozapine: monitor the CBC

Acute dystonia
High-potency typicals, within hours to days: treat with **IV benztropine**

Long-acting injection
Haloperidol, fluphenazine, risperidone, paliperidone, aripiprazole

Two emergencies to tell apart

	Cause	Findings	Treatment
Neuroleptic malignant syndrome	Dopamine antagonists, often high-potency, high-dose or rapidly raised typicals	Fever, labile BP and pulse, sweating, altered mental status, **lead-pipe rigidity**; high CK, kidney injury; ~20% die	Stop the drug; supportive care; dantrolene, bromocriptine; ECT
Serotonin syndrome	Several serotonergic drugs, or an SSRI plus an interacting drug; within ~48 h	Tremor, sweating, hypertension, **hyperreflexia and clonus**	Stop the drugs; supportive care, IV fluids; cyproheptadine, benzodiazepines

Ask: **"Which side effects can this patient least afford?"** Diabetes or obesity → avoid clozapine and olanzapine. Prolactin problems → avoid risperidone. Poor adherence → long-acting injection. Acute agitation → safety first: benzodiazepine or a sedating antipsychotic.

Sources: E25 Psychotic Disorders and Pharmacology slides; Psychiatry Exam Review deck (NMS vs serotonin syndrome, EPS); Mood Disorders slides (ECT for NMS). Atypicals were promoted as safer and more effective than typicals; that has not held up. Treatment of schizophrenia is usually lifelong.

Schizophrenia: course, prognosis and side effects over time

Onset is usually late teens to early 30s, earlier in men (peak early to mid 20s) than women (late 20s). The course varies: roughly a third each do well, partly improve, or stay severely affected.

Good and poor prognostic factors

Factor	Good prognosis	Poor prognosis
Sex	**Female**	Male
Onset	Abrupt, acute; later	Insidious; early
Main symptoms	Positive	Negative
Mood symptoms	Present	Absent
Premorbid functioning	Good	Poor
Social support	Good	Poor
Relapses	Few	Frequent
Other		Family history; substance use

Positive vs negative symptoms: positive symptoms are added (hallucinations, delusions, disorganized speech and behavior; mesolimbic). Negative symptoms are missing (flat affect, anhedonia, apathy, alogia, asociality; mesocortical).

Mortality: 5–6% die by suicide and 20–50% attempt. Life expectancy is shorter from medical illness, metabolic effects of medication and poor access to care. Substance use (especially tobacco) and anxiety disorders are common.

Antipsychotic side effects on a timeline

Effect	When	What it looks like	Response
Acute dystonia	Hours to days	Sustained contractions of tongue, face, neck; eyes forced upward; jaw clenched	**IV benztropine**
Acute parkinsonism	Hours to days	Bradykinesia, resting tremor, rigidity, shuffling gait	Highest risk with high-potency typicals
Akathisia	Early	Can’t sit still; inner restlessness and distress	Common with aripiprazole
Neuroleptic malignant syndrome	Days	Fever, rigidity, autonomic instability, confusion; very high CK	Stop drug; dantrolene, bromocriptine
Metabolic syndrome	Months	Weight gain, diabetes, dyslipidemia	Monitor weight, glucose, A1c, lipids
Tardive dyskinesia	After prolonged use	Grimacing, tongue movements, lip smacking, blinking	High-potency typicals carry the most risk

Clozapine
Agranulocytosis (strict blood monitoring), seizures, myocarditis, weight gain

Quetiapine
Least EPS: preferred for psychosis in Parkinson disease; sedation, orthostasis

Ziprasidone
Must be taken with food to be absorbed

Low-potency typicals
Anticholinergic effects, orthostasis, weight gain, sedation

For prognosis, ask: **“How did it start, and what was the person like before?”** A sudden onset later in life, in a woman with good functioning, mood symptoms and support, predicts a better course than an insidious onset in an adolescent male with negative symptoms.

Sources: E25 Psychotic Disorders and Pharmacology slides (course, prognosis, mortality); Psychiatry Exam Review deck (prognostic factors table; side effects of FGAs and SGAs); Neuropsychiatry slides (NMS).

MBB PSYCHIATRY · PERSONALITY DISORDERS

The ten personality disorders at a glance

Personality disorders are lifelong patterns, not episodes. Name the cluster first, then the core pattern that ties the traits together.

Disorder	Core pattern	Look for	Easily confused with
CLUSTER A · ODD OR ECCENTRIC			
Paranoid	Pervasive distrust and suspicion	Assumes others mean harm, bears grudges, reads threats into remarks, reluctant to confide	Delusional disorder (a fixed delusion)
Schizoid	Detachment from relationships	Prefers to be alone, little interest in intimacy, restricted emotion, indifferent to praise or criticism	Avoidant (wants relationships but fears rejection)
Schizotypal	Odd thinking and behavior	Magical thinking, ideas of reference, unusual perceptions, eccentric speech or dress, social anxiety	Schizophrenia (sustained psychosis)
CLUSTER B · DRAMATIC, EMOTIONAL OR ERRATIC			
Antisocial	Disregard for others' rights	Deceit, impulsivity, aggression, no remorse; **age 18 or older with conduct disorder before 15**	Substance use (behavior only while using)
Borderline	Instability of relationships, self-image and mood	Fear of abandonment, idealizing then devaluing, impulsivity, recurrent self-harm, chronic emptiness, intense anger, stress-related paranoia	Bipolar disorder (distinct episodes lasting days or longer)
Histrionic	Attention seeking and excessive emotion	Needs to be the center of attention, dramatic, shallow or rapidly shifting emotion, seductive behavior	Narcissistic (seeks admiration, not just attention)
Narcissistic	Grandiosity, need for admiration, lack of empathy	Entitlement, exploits others, envy, arrogance; fragile self-esteem underneath	Mania (episodic grandiosity)
CLUSTER C · ANXIOUS OR FEARFUL			
Avoidant	Feelings of inadequacy and fear of rejection	Wants relationships but avoids them unless sure of being liked; hypersensitive to criticism	Social anxiety disorder (often coexists)
Dependent	Excessive need to be taken care of	Can't make everyday decisions alone, fears being left to care for self, seeks a new relationship urgently when one ends	Borderline (reacts to abandonment with rage, not submission)
Obsessive-compulsive (OCPD)	Perfectionism, order and control	Rules and lists, can't delegate, rigid, overly devoted to work; sees these traits as correct	OCD (unwanted intrusive thoughts and rituals)

Remember the clusters: A is weird, B is wild, C is worried.

Usually ego-syntonic: the person sees the traits as part of who they are, and others are often more distressed than the patient.

All personality disorders are enduring, inflexible patterns across many situations, starting by adolescence or early adulthood.

Treatment is mainly psychotherapy. Dialectical behavior therapy for borderline personality disorder has the strongest evidence; medications target specific symptoms only.

Criteria follow DSM-5-TR. Diagnose a personality disorder only when the pattern isn't better explained by another mental disorder, a substance or a medical condition.

Personality disorders and their look-alikes

Each personality disorder has a symptom-based disorder that looks like it. The time course (lifelong pattern vs. episode) and the person's attitude toward the behavior usually decide.

Personality disorder	Look-alike	Points to the personality disorder	Points to the look-alike
Borderline	Bipolar disorder	Mood shifts over hours, triggered by relationships; chronic emptiness and fear of abandonment	Distinct episodes lasting days to weeks; decreased need for sleep; baseline between episodes
Avoidant	Social anxiety disorder	Pervasive sense of inadequacy across all of life since early adulthood	Fear limited to certain social or performance situations
Schizoid	Avoidant	Doesn't want relationships and is content alone	Wants relationships but stays away for fear of rejection
Schizotypal	Schizophrenia	Odd beliefs and perceptions without sustained psychosis	Delusions or hallucinations for 1 month or more; functional decline
Paranoid	Delusional disorder	General suspiciousness of many people	One fixed false belief held with full conviction
Obsessive-compulsive (OCPD)	OCD	Lifelong perfectionism the person sees as correct	Unwanted intrusive thoughts and distressing rituals
Antisocial	Conduct disorder	Age 18 or older, with conduct disorder before 15	Under 18
Narcissistic	Mania	Stable, lifelong grandiosity and entitlement	Grandiosity during an episode, with decreased sleep and pressured speech
Dependent	Borderline	Clings and submits to avoid being left	Reacts to feared abandonment with anger, impulsivity or self-harm
Histrionic	Narcissistic	Wants attention of any kind; emotions shallow and shifting	Wants admiration; feels superior and entitled
Any personality disorder	Personality change due to a medical condition	Present since adolescence or early adulthood	New change in personality after a brain injury, stroke or other illness

Ask **"When did this start, and is it always this way?"** A personality disorder has been present since adolescence or early adulthood and shows up across work, home and relationships. A pattern that comes and goes, or began recently, points to an episodic disorder or a medical cause.

Criteria follow DSM-5-TR. A personality disorder and its look-alike can coexist; diagnose both when both are met.

Treating personality disorders

Psychotherapy is the treatment. Medications are prescribed often, but the evidence for them is weak, so they target specific symptoms for limited periods.

Psychotherapy comes first

Disorder	Therapies with evidence
Borderline	**Dialectical behavior therapy** (strongest evidence: fewer self-harm episodes, suicide attempts and hospitalizations); mentalization-based treatment; transference-focused psychotherapy; schema-focused therapy; general psychiatric management
Other personality disorders	Psychodynamic, supportive, DBT. Evidence is limited to small open-label trials and case studies, with positive findings

Borderline personality disorder is treatable. DBT teaches four skill areas: mindfulness, emotion regulation, distress tolerance and interpersonal effectiveness, in individual therapy, a skills group and phone coaching for about a year.

Watch your own reactions. These patients can stir strong feelings in clinicians (countertransference). Noticing them protects you from inappropriate relationships and care outside your usual standard.

Medications: common, but weakly supported

THE EVIDENCE

No consistent support

- No consistent evidence supports medication for **any** personality disorder
- UK NICE guidance: use only in emergencies, for limited periods

THE PRACTICE

Prescribed anyway

- 70–80% of outpatients with personality disorders take antidepressants
- 35–60% take antipsychotics

Target symptom (borderline)	Medication sometimes used
Impulsivity	SSRIs
Mood lability, anger, chronic suicidal thoughts	Mood stabilizers (lithium), antipsychotics

Ask: **“Which symptom is this drug for, and for how long?”** Treat a coexisting disorder (major depression, bipolar disorder, PTSD) on its own terms. For the personality disorder itself, refer for psychotherapy.

Sources: E25 Personality Disorders slides (therapy and medication evidence); Psychotherapy slides; Psychiatry Exam Review deck (DBT; target-symptom medications).

MBB PSYCHIATRY · NEUROPSYCHIATRY

Delirium and catatonia

Delirium is acute brain failure: a disturbance of attention and awareness, plus another brain function, that develops over hours to days and fluctuates within a single day. It always has a cause.

	Delirium	Dementia	Depression
Onset	**Hours to days**	Months to years	Weeks
Course	**Waxes and wanes** within a day	Slowly progressive	Persistent low mood
Attention, arousal	**Impaired; drowsy or agitated**	Intact until late	Mostly intact; slowed
Other features	Disorientation, memory and language problems, **visual hallucinations**, sleep–wake reversal	Memory loss first	Sadness, anhedonia, guilt

HYPOACTIVE

Slow, spacey, vague, somnolent. Easily mistaken for depression or "tiredness"

TEST ATTENTION

2–3 step commands; WORLD, days or months backward; serial 7s

HYPERACTIVE

Restless, perseverative, vigilant, agitated

Causes: predisposing × precipitating

PREDISPOSING

Older age, dementia, frailty, poor vision or hearing, chronic illness, poor nutrition, **prior delirium**, alcohol use disorder

PRECIPITATING (AEIOU TIPSS)

Alcohol or drugs (intoxication, withdrawal), electrolytes (esp. sodium), iatrogenic, oxygen, uremia or liver failure, trauma, infection, poisons, seizures, stroke; poor sleep

Deliriogenic drugs: anticholinergics (diphenhydramine, hydroxyzine, oxybutynin, amitriptyline, paroxetine, benztropine, scopolamine, clozapine, olanzapine), benzodiazepines, opioids, muscle relaxants, corticosteroids.

Prevent and manage

Do	Don't
Treat the cause: infection, electrolytes, oxygen, organ failure	Order deliriogenic medications
Orient often; glasses and hearing aids	**Wake patients repeatedly at night**
Get out of bed; hydration, nutrition	Leave urinary catheters in
Edit the medication list	Keep patients in bed for days

> **Medications have a limited role.** Benzodiazepines only for alcohol or sedative withdrawal. Antipsychotics (haloperidol, quetiapine) treat agitation and psychosis, not the delirium itself.

Catatonia

- **Signs:** posturing, waxy flexibility (catalepsy), mutism, echolalia, echopraxia, negativism
- **Causes:** severe **mood disorders** (depression, bipolar disorder), psychotic disorders, medical conditions. Not specific to schizophrenia
- **Treatment:** high-dose benzodiazepines (**IV lorazepam**, a GABA-A positive allosteric modulator) bring rapid improvement; **ECT** is the gold standard if it persists

	Serotonin syndrome	NMS
Onset	Fast, ~24 h	Slow, several days
Muscles	Tremor, hyperreflexia, clonus	Rigidity, reduced reflexes, high CK
Resolves	~24 h	Days to weeks

Any sudden change in behavior over a day or two: think **delirium first**. Ask: **"Can the patient hold attention, and does it fluctuate?"** Then look for every cause, especially medications.

Sources: E25 Delirium slides; Neuropsychiatry and Neuromodulation slides (catatonia, SS vs NMS); Mental Status Exam slides; Psychiatry Exam Review deck (delirium risk factors, catatonia). Dementia and depression columns summarize what the delirium lecture says delirium is mistaken for.

Intoxication and withdrawal

Withdrawal usually looks like the opposite of intoxication. Pupils, temperature and vital signs separate most of them. Alcohol and benzodiazepine withdrawal can kill; opioid withdrawal is miserable but not life-threatening.

Substance	Mechanism	Intoxication	Withdrawal	Danger
Alcohol	Activates GABA-A; inhibits glutamate (NMDA) and calcium channels	Potent CNS depressant: sedation; lethal respiratory depression with benzodiazepines	Starts 6–8 h after the last drink, peaks 24–48 h: anxiety, tremor, sweating, GI upset, insomnia, high BP and pulse. **Seizures** at 24–48 h; **delirium tremens** at 48–96 h	**Withdrawal can kill**: delirium tremens (confusion plus autonomic instability) up to 25% mortality untreated
Benzodiazepines	Potentiate GABA; cross-tolerant with alcohol	Sedation; respiratory depression, lethal with alcohol	Like alcohol, including seizures	**Withdrawal can kill**
Opioids	Mu agonists (full: heroin, oxycodone, fentanyl, methadone; partial: buprenorphine)	**Pinpoint pupils**, euphoria, constipation, sedation, **respiratory depression**, coma	**Dilated pupils**, yawning, rhinorrhea, **piloerection**, sweating, hot/cold flashes, aches, nausea, vomiting, cramps, diarrhea, high BP and pulse	**Overdose kills**; withdrawal does not
Cocaine, methamphetamine	Raise synaptic dopamine (by different mechanisms)	Dilated pupils, tachycardia, hypertension, hyperthermia, sweating, anorexia, insomnia; euphoria, paranoia, hallucinations, aggression	"Crash": extreme fatigue, depression, dysphoria, increased appetite, suicidal thoughts	Vasoconstriction → MI, stroke; arrhythmias, seizures
Cannabis	THC: CB1/CB2 agonist (CBD: antagonist)	Tachycardia, **red eyes** (conjunctival injection), dry mouth, increased appetite, **slowed time**; anxiety, paranoia, psychosis	Craving, irritability, anxiety, insomnia, restlessness, depressed mood	Heavy teen use raises later psychosis risk (COMT variants)
MDMA (ecstasy, "Molly")	Stimulant and hallucinogen: releases serotonin, dopamine, norepinephrine	Euphoria, emotional warmth; **hyperthermia**, dehydration, **bruxism** (jaw clenching), **hyponatremia**, serotonin syndrome		Hyperthermia, organ failure, rhabdomyolysis, seizures; "Molly" contents unknown
PCP	NMDA antagonist	Violence, ataxia, hypertension, tachycardia, **rotary nystagmus**, hallucinations, synesthesia, depersonalization		Psychosis with positive and negative symptoms
Synthetic drugs	Cannabinoids ("spice"); cathinones ("bath salts") raise NE, DA, 5-HT	Spice: psychosis, agitation, tachycardia, seizures. Bath salts: agitation, sweating, palpitations, hallucinations, seizures		Contents change constantly
Nicotine	Nicotinic acetylcholine receptors	Restlessness, insomnia, anxiety	Craving, dysphoria, anxiety, poor concentration, increased appetite, irritability, insomnia	Leading cause of preventable death

Quick clues

Pinpoint pupils
Opioid intoxication

Dilated pupils
Stimulants; opioid withdrawal

Red eyes, slow time
Cannabis

Rotary nystagmus, violence
PCP

Hyperthermia, bruxism, low Na
MDMA

Tremor, sweats, rising BP
Alcohol or benzodiazepine withdrawal

Ask: **"Which withdrawal can kill?"** Alcohol and benzodiazepines: seizures and delirium tremens. For opioids it is the **overdose** that kills, through respiratory depression.

Sources: E25 Substance Use Disorders slides (Intro, Alcohol, Opioids); Psychiatry Exam Review deck (intoxication and withdrawal presentations); Psychotic Disorders slides (PCP).

MBB PSYCHIATRY · SUBSTANCE USE · TREATMENT

Treating substance use disorders

A substance use disorder is a problematic pattern with 2 or more of 11 criteria in 12 months. Tolerance and withdrawal from medication taken as prescribed don't count: physiologic dependence is not addiction.

Alcohol: withdrawal, then relapse prevention

EMERGENCY DEPARTMENT

- IV fluids with **thiamine** and folate ("banana bag")
- Lorazepam or another benzodiazepine
- Nutrition; consider admission

ADMIT IF

- High CIWA score, abnormal vitals
- Past seizures or delirium tremens
- Medical or unstable psychiatric problems
- Outpatient detox has failed

INPATIENT DETOX

- **Everyone gets a CIWA** (≤9 mild, 10–18 moderate, >18 severe)
- **Scheduled** benzodiazepines if past seizures or DTs
- Thiamine: prevent Wernicke-Korsakoff

Drug	Mechanism	Best for	Watch for
Naltrexone	Opioid receptor antagonist; cuts craving and reward	Cravings, family history; daily pill or monthly injection	Nausea, dizziness; **precipitates withdrawal if taking opioids**; opioid pain relief won't work
Acamprosate	GABA-like; may inhibit glutamate	**Liver disease**	Diarrhea; three times daily; caution in kidney disease
Disulfiram	Blocks **aldehyde dehydrogenase**, so acetaldehyde builds up	Highly motivated, monitored patients; patients on opioids	With alcohol: flushing, headache, vomiting, palpitations. Liver failure (check LFTs); avoid in pregnancy, psychosis, heart disease

Stimulants and nicotine

Cocaine, methamphetamine	No FDA-approved medication. **Contingency management** (rewarding abstinence) has the most evidence; overdose care is supportive
Varenicline	α4β2 nicotinic partial agonist; mimics nicotine and blocks withdrawal. GI upset, psychiatric changes
Bupropion	Reduces nicotine withdrawal. Tremor, anxiety; lowers the seizure threshold
Nicotine replacement	Patch for a steady level plus gum or lozenge for cravings

Opioids: overdose and maintenance

OVERDOSE

- Airway first, then **naloxone** (IV, IM, nasal): a short-acting competitive antagonist that **displaces the opioid** from the mu receptor
- Naloxone can wear off before a long-acting opioid does: **observe for hours**

Drug	Key points
Methadone full mu agonist	Most studied; daily at an opioid treatment program. **QTc prolongation**; constipation, weight gain, many drug interactions. No precipitated withdrawal
Buprenorphine mu partial agonist	High affinity; **ceiling on respiratory depression**, so safer. Sublingual or monthly injection. **Start only once in withdrawal**, or it precipitates withdrawal. In pregnancy: methadone or buprenorphine, and **buprenorphine is now first line**
Naltrexone mu antagonist	Monthly injection. Harder to start (must be opioid-free), but similar relapse prevention once started

> **Methadone and buprenorphine save lives:** both lower morbidity and mortality and keep patients in treatment.

> **Before buprenorphine, ask: "When did you last use?"** Too soon after a full agonist, it strips the receptor and causes sudden withdrawal.

Match the drug to the goal. **Alcohol:** reduce craving → naltrexone; liver disease → acamprosate; motivated and supervised → disulfiram. **Opioids:** reverse overdose → naloxone; maintenance → methadone or buprenorphine. **Nicotine:** varenicline, bupropion or combination replacement.

Sources: E25 Substance Use Disorders slides (Intro, Alcohol, Opioids) and notes. Substance Use Disorder course pack (pregnancy).

MBB PSYCHIATRY · PSYCHOTHERAPY

Telling the psychotherapies apart

Name a therapy by what happens in the room and how it is set up. A trusting therapeutic alliance (interest, empathy, understanding, hope) predicts outcome in all of them.

Therapy	Core idea	What happens in session	Format and length	Best for
Supportive	Support here-and-now coping; grew out of psychodynamic therapy	Listening; reflecting, naming, normalizing and reframing feelings; encouragement and advice	Weekly or less; part of most medication visits	Hard situations and transitions (a new diagnosis, a loss); any diagnosis, or none
Psychodynamic / psychoanalytic	Unconscious conflicts from childhood and early relationships shape feelings and relationships	Speaking freely: whatever comes to mind, dreams, fantasies, feelings about the therapist; exploring **transference** and defenses	Weekly or more (psychoanalysis 3–5 times a week); **months to years**	Depression, anxiety, relationship problems, personality disorders
Cognitive behavioral (CBT)	Thoughts, feelings and behaviors drive each other (the CBT triangle)	Catch automatic thoughts, name the distortion, weigh the evidence (**cognitive restructuring**); change behaviors; homework	Manual-guided, weekly; about 8–16 sessions over weeks to months	Depression, GAD, panic, social anxiety, phobias; psychosis, substance use; insomnia (CBT-I)
Behavioral activation	A part of CBT: rewarding activity lifts mood, which makes more activity likely (positive reinforcement)	Set specific, achievable activity goals for the week; review them next session	Structured, within CBT	Depression, including perinatal depression
Dialectical behavior (DBT)	Skills for emotion dysregulation; developed for borderline personality disorder	Mindfulness, emotion regulation, distress tolerance, interpersonal effectiveness	Individual therapy + skills group + phone coaching; **about 1 year**	Borderline personality disorder; adolescents who self-injure; depression, substance use, eating disorders
Motivational interviewing	Explore and resolve ambivalence; change the patient chooses lasts	Ask what the behavior gives them and what worries them; reflect both sides; no lecturing	Often brief; primary care and mental health	Behavior change: alcohol and drugs, smoking, medication adherence, exercise, diabetes
Interpersonal (IPT)	Problems in relationships drive depression	Map the “cast of characters”; work on conflicts, role transitions, grief	Listed for reference in this course	Depression, perinatal depression, grief

Therapy, medication or both?

THERAPY FIRST LINE

Mild depression, mild anxiety, specific phobias, personality disorders, eating disorders, PTSD, insomnia (CBT-I)

BOTH ARE COMMON

Moderate to severe depression and anxiety, OCD

MEDICATION FIRST LINE

Schizophrenia, bipolar disorder (therapy still helps, e.g. CBT reduces relapse in schizophrenia)

Does it work?

Effect size 0.73–0.85 for psychotherapy overall, compared with about 0.30 for antidepressants and 0.56 for antihypertensives. Unlike medication, the benefit usually lasts after treatment ends.

Teletherapy seems about as effective, with caveats: the alliance may build more slowly, and some therapists are less skillful on video.

Ask: **“What does the therapist actually do in session, and how often, for how long?”** Distortions and homework → CBT. Activity goals → behavioral activation. Free association and feelings about the therapist, for years → psychodynamic. Skills group for self-harm → DBT. Weighing pros and cons of a habit → motivational interviewing. Comfort and coping through a crisis → supportive.

Sources: E25 Psychotherapy slides and 2026 course pack; Psychiatry Exam Review deck (comparing psychotherapies). Most therapists are eclectic and mix approaches.

Matching the therapy to the problem

Several disorders have a specific, named therapy. Most of the exposure-based ones are forms of CBT; what differs is what the patient faces and how fast.

Problem	Therapy	What it involves
Panic disorder	CBT	Recreate the body sensations (e.g. jogging in place), correct catastrophic thoughts, reduce escape and avoidance
GAD	CBT; mindfulness, relaxation training	Self-monitor worry, challenge automatic thoughts, imaginal exposure, coping skills
Agoraphobia	Exposure therapy (CBT if panic too)	Exposure "in the field" to the avoided places
Specific phobia	Systematic desensitization or flooding	Relaxation first, then a **gradual** fear hierarchy (desensitization) or the most feared level **all at once** (flooding)
OCD	Exposure and response prevention (ERP)	Face the trigger, **don't perform the compulsion**; climb a hierarchy. Equal or superior to medication
PTSD	Prolonged exposure (PE) or cognitive processing therapy (CPT)	PE: approach avoided memories and situations safely. CPT: write and reread an account of the trauma, questioning its meanings
Trichotillomania, excoriation	Habit reversal training	Self-monitoring, awareness training, stimulus control, a **competing response**
Hoarding; body dysmorphic disorder	CBT for hoarding; CBT	Specialized CBT
Depression	CBT, behavioral activation, IPT, ACT, psychodynamic, supportive	Mild depression: therapy alone is first line
Borderline personality disorder	DBT; mentalization-based, transference-focused, psychodynamic, general psychiatric management	DBT reduces self-harm, suicide attempts and hospitalizations
Psychosis	CBT, supportive	CBT can reduce relapse in schizophrenia
Ambivalence about a habit	Motivational interviewing	Meet the patient at their stage of change

Supportive interventions anyone can use

Patient: overwhelmed by a move and a new job.

Reflecting	Helps the patient feel heard: "You've faced a lot of stress this month."
Naming	Gives mastery over the feeling: "It sounds like you feel overwhelmed."
Normalizing	Reduces guilt and shame: "Anyone facing this would feel overwhelmed."
Reframing	Offers a more hopeful view: "Getting through this has brought you closer to your partner."

Feelings in the room

PATIENT → CLINICIAN

Transference

The patient's feelings about the therapist, from the therapist's real traits and from important past relationships.

CLINICIAN → PATIENT

Countertransference

The clinician's feelings about the patient, from the interaction and their own past. Happens in every specialty; unexamined, it can bias care.

Example: a patient who sees the doctor as an authority apologizes again and again for being late; the doctor grows exasperated. Both reactions show how the patient relates to others.

> **Defenses** (denial, projection, splitting, sublimation and others) are unconscious ways of coping with painful feelings. High-yield for boards; the course pack says they are **not on this course's final exam**.

Ask: **"What exactly does the patient need to face, or to change?"** A feared object → desensitization or flooding. A compulsion → ERP. Trauma reminders or beliefs → PE or CPT. A body-focused habit → habit reversal. Self-harm with stormy relationships → DBT.

Sources: E25 Psychotherapy slides and 2026 course pack; E25 Anxiety Disorders and Pharmacology slides; Personality Disorders slides; Psychiatry Exam Review deck.